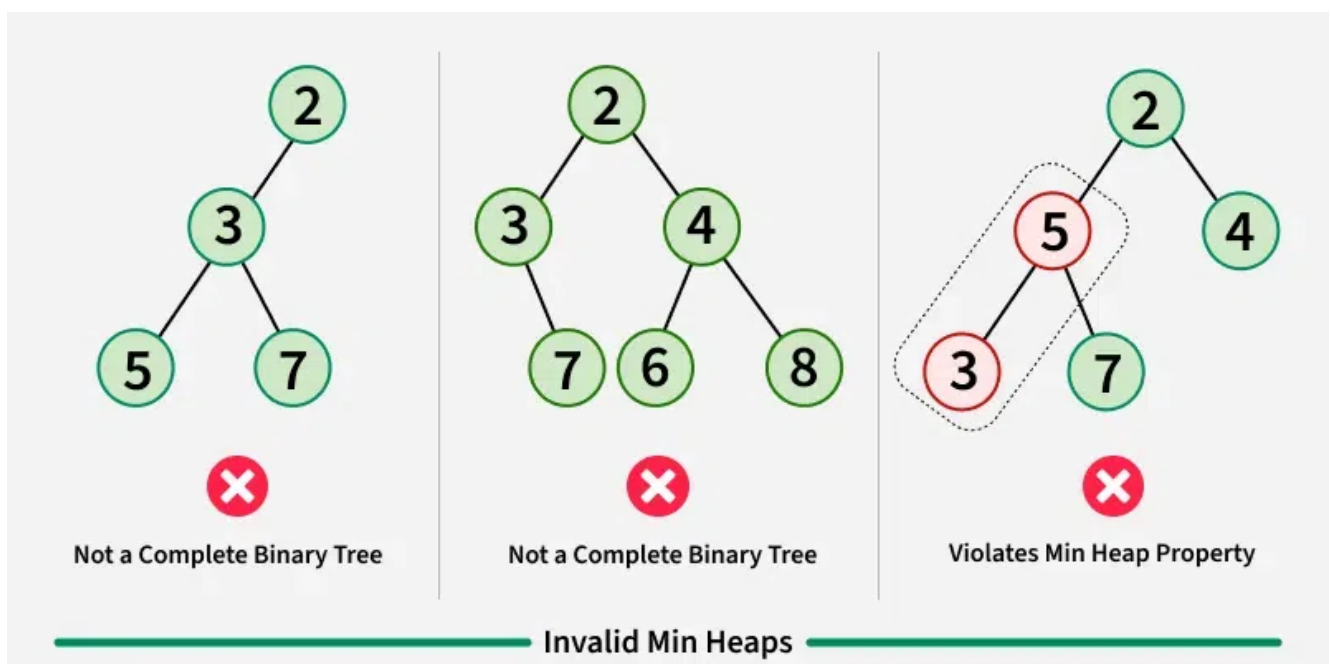
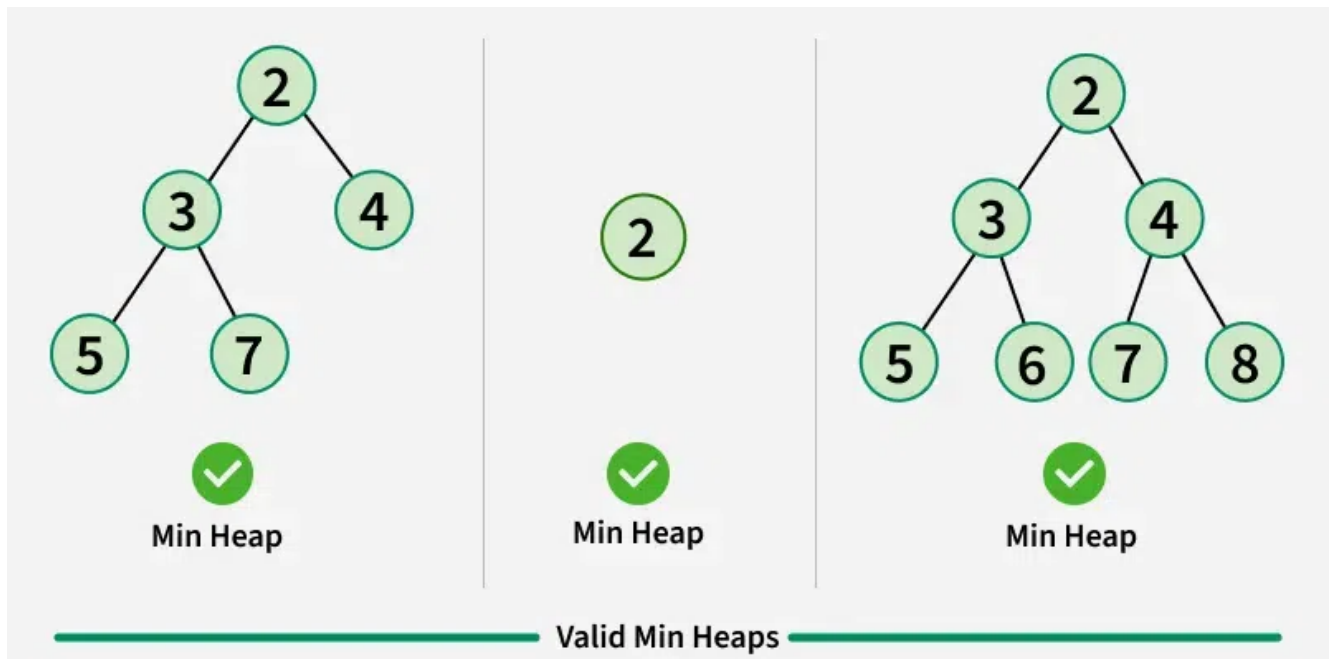


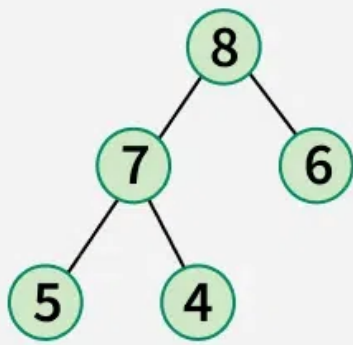
Heap Data Structure

Last Updated : 26 Apr, 2025



A **Heap** is a complete binary tree data structure that satisfies the heap property: for every node, the value of its children is greater than or equal to its own value. Heaps are usually used to implement priority queues, where the smallest (or largest) element is always at the root of the tree.

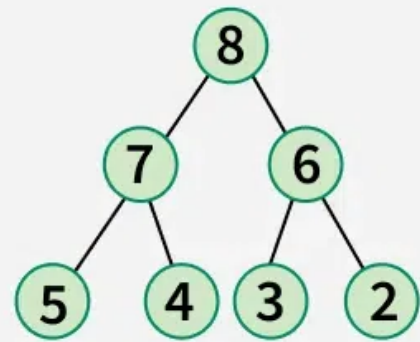




Max Heap

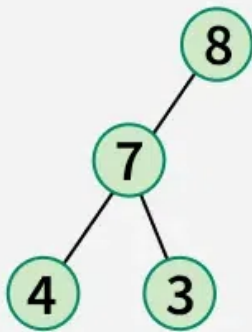


Max Heap

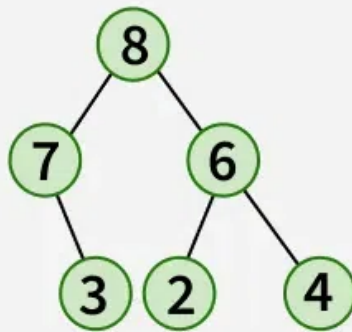


Max Heap

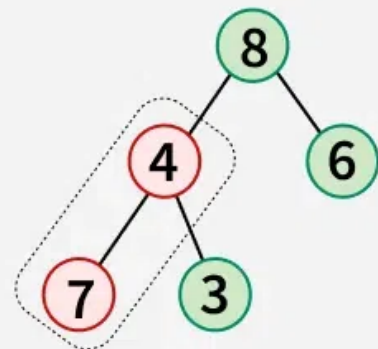
Valid Max Heaps



Not a Complete Binary Tree



Not a Complete Binary Tree



Violates Max Heap Property

Invalid Max Heaps

Basics

- [Binary Heap](#)
- [Applications of Heap Data Structure](#)

Library Implementations

- [Heap in C++ STL](#)
- [priority_queue in C++](#)
- [PriorityQueue in Java](#)
- [PriorityQueue in C#](#)
- [heapq in Python](#)
- [Heap in JavaScript](#)

Easy Problems

- [Check if an array is Heap?](#)
- [K'th Smallest & K'th Largest](#)
- [Heap Sort](#)
- [Heap Sort for Decreasing Order](#)
- [Height of a Heap with N Nodes](#)

Medium Problems

- [Nearly Sorted Array](#)
- [K Largest Elements & K Smallest](#)
- [k Closest Elements](#)
- [Max diff between two m Sized Subsets](#)
- [Check if Binary Tree is Heap](#)
- [Huffman Coding](#)
- [Nodes less than a value in a Min Heap.](#)
- [Connect n ropes with minimum cost](#)
- [Maximum distincts after removing k elements](#)
- [K maximum sum combinations from two arrays](#)
- [Median in a stream](#)
- [K'th largest in a stream](#)
- [Largest triplet product in a stream](#)
- [k most frequent](#)
- [Sort by Frequency](#)
- [Min Heap to Max Heap](#)
- [Check for Min-Heap from Level Order](#)

Hard Problems

- [The Skyline Problem](#)
- [K-th Largest Sum Subarray](#)
- [Find K Closest Points to the Origin](#)
- [Merge k sorted arrays](#)
- [Merge K Sorted Linked Lists](#)
- [Range Covering K Sorted Lists](#)

- [Prim's Minimum Spanning Tree](#)
- [Dijkstra's Shortest Path](#)
- [Sort numbers stored on different machines](#)
- [Smallest Derangement of Sequence](#)
- [Largest Derangement of a Sequence](#)
- [Convert BST to Min Heap](#)
- [Merge two binary Max Heaps](#)
- [Minimum product of k integers](#)
- [Leaf starting point in a Binary Heap](#)
- [Rearrange characters such that no two same adjacent](#)
- [Sum of all between k1'th and k2'th smallest](#)
- [Min sum of two formed from given digits](#)
- [A data structure with min and max](#)

Other Types of Heaps

- [Binomial Heap](#)
- [Fibonacci Heap](#)
- [Leftist Heap](#)
- [K-ary Heap](#)

Quick Links:

- [Practice Problems on Heap](#)
- [Quizzes on Heap](#)

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Advantages and Disadvantages of Heap

Advantages of Heap Data Structure Time Efficient: Heaps have an average time complexity of $O(\log n)$ for inserting and deleting elements, making them efficient for large datasets. We can convert any array to a...

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Time Complexity of building a heap

Consider the following algorithm for building a Heap of an input array A. A quick look over the above implementation suggests that the running time is $O(n * \lg(n))$ since each call to Heapify costs $O(\lg(n))$ an...

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Applications of Heap Data Structure

Heap Data Structure is generally taught with Heapsort. Heapsort algorithm has limited uses because Quicksort is better in practice. Nevertheless, the Heap data structure itself is enormously used. Priority...

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Comparison between Heap and Tree

What is Heap? A Heap is a special Tree-based data structure in which the tree is a complete binary tree. Types of Heap Data Structure: Generally, Heaps can be of two types: Max-Heap: In a Max-Heap the key...

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When building a Heap, is the structure of Heap unique?

What is Heap? A heap is a tree based data structure where the tree is a complete binary tree that maintains the property that either the children of a node are less than itself (max heap) or the children a...

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Some other type of Heap



Easy problems on Heap

